

Course Project Proposal

Predicting HSBC Weekly Stock Price Using Regression

Data Science with Foundation Model | Spring 2026

Research Question

Can we predict HSBC Holdings' weekly closing stock price using publicly available market data and machine learning regression techniques? Specifically, we aim to build a regression model that takes this week's market features as input and outputs a prediction for next week's HSBC closing price. Along the way, we will investigate which categories of features — HSBC's own price history, banking sector peers, macroeconomic indicators, or broader market sentiment — contribute most to predictive accuracy.

Why This Project?

Stock price prediction is one of the most compelling applications of data science — it sits at the intersection of statistics, economics, and machine learning. HSBC is a particularly interesting case study because it is a global bank listed in both London and Hong Kong, with revenue streams spanning multiple currencies and economies. This means its stock price is influenced by a rich set of factors: interest rates, exchange rates, peer bank performance, and global market sentiment. The project provides students with hands-on experience across the full data science pipeline — from data collection and feature engineering to model building and critical evaluation — while engaging with the fundamental question of market predictability.

Data

All data will be collected from Yahoo Finance using the open-source *yfinance* Python library at no cost. We will gather approximately five years of weekly data (2019–2025, ~250+ observations) across four feature buckets: **(1) HSBC itself** — closing price, volume, weekly high/low, moving averages, momentum, and volatility indicators; **(2) Banking sector peers** — JPMorgan, Citigroup, Standard Chartered, and the XLF financial sector ETF; **(3) Macroeconomic environment** — GBP/USD, HKD/USD, CNY/USD exchange rates and the US 10-year Treasury yield; **(4) Market sentiment** — S&P 500, FTSE 100, Hang Seng Index, VIX, and gold futures. The target variable is next week's HSBC closing price.

Modeling Approach

We will apply a progression of regression techniques covered in the course. We begin with **linear regression** as a baseline to understand the linear relationship between features and price. We will then explore **ridge and lasso regression** for regularization and feature selection, **decision tree and random forest regression** to capture non-linear patterns, and potentially **gradient boosting (XGBoost)** for improved predictive power. Model performance will be evaluated using RMSE, MAE, and R^2 on a held-out test set, with careful attention to avoiding lookahead bias. We will compare all models against a naive baseline (predicting that next week's price equals this week's price) to honestly assess whether our models add value. Foundation models (Claude) are used throughout for ideation, code assistance, and analysis — but the predictive models themselves are built using classical ML techniques.

Six-Week Timeline

Weeks 1–2: Data collection, cleaning, and exploratory data analysis (EDA).

Week 3: Feature engineering — technical indicators, cross-asset features, and correlation analysis.

Week 4: Model building — train baseline and advanced regression models.

Week 5: Model evaluation, comparison, and interpretation of feature importance.

Week 6: Final report and presentation — findings, lessons learned, and reflections on market predictability.